BUILD-A-THON 1.0 · THEME 03 · ALSO COVERS THEMES 01, 02, 04

StageCraft

The audience is part of the show.

Live events for IITM BS houses: interaction, brackets, talent hunts, a broadcast overlay and verifiable certificates — in one app.

stagecraft-xi.vercel.app github.com/lakshanmuruganandam/stagecraft

DEMO LOGINS · PASSWORD HOUSE-CUP-2026

`admin@saranda.demo`
Admin — users, roles, audit, broadcast

`organizer@saranda.demo`
Organizer — control room, wrap-up

`member@saranda.demo`
Member — ask, vote, react, predict

Demo mode: the avatar menu switches personas instantly.

SCROLL · ← → · SPACE TO NAVIGATE · F FULLSCREEN

01

THE PROBLEM

House events today are duct-taped together.

Scattered tools

A YouTube link, a Google Form for questions, a WhatsApp group for everything else.

Passive audiences

Hundreds watch, few take part. No live Q&A, no polls, no stake in the result.

Juggling organizers

Scores in a sheet, votes counted by hand, five tabs open while running the show.

Unverifiable certificates

PDFs made days later in Canva; anyone can edit one, nobody can check one.

Result: low turnout, lost engagement data, and house points nobody trusts.

THE PRODUCT

One live room. Every part of the show.

Event room
Stream + live chat with slow mode, pins and moderation.

Q&A · polls · reactions
Upvoted questions, live polls, floating house-coloured reactions.

Brackets + predictions
Esports brackets; members predict winners before each match.

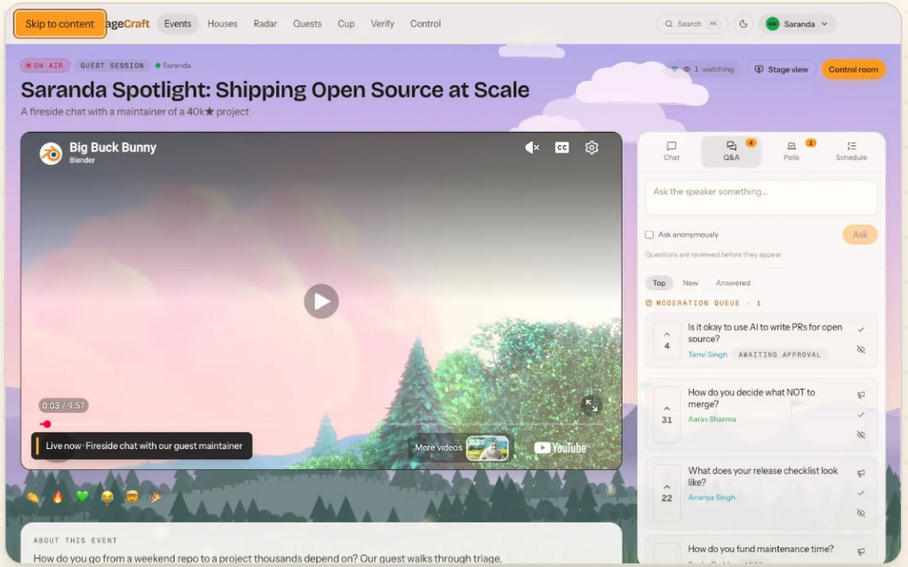
Sealed talent-hunt votes
Votes hidden until the reveal, judges weighted separately.

Control room
Run the whole show from one screen: segments, scenes, scores.

OBS overlay + signed QR
Stage overlay with a rotating HMAC-signed check-in code.

One-click certificates
Ed25519-signed, verifiable, plus house points on wrap-up.

House platform
Hubs, calendar radar, quests & badges (Themes 01, 02, 04).



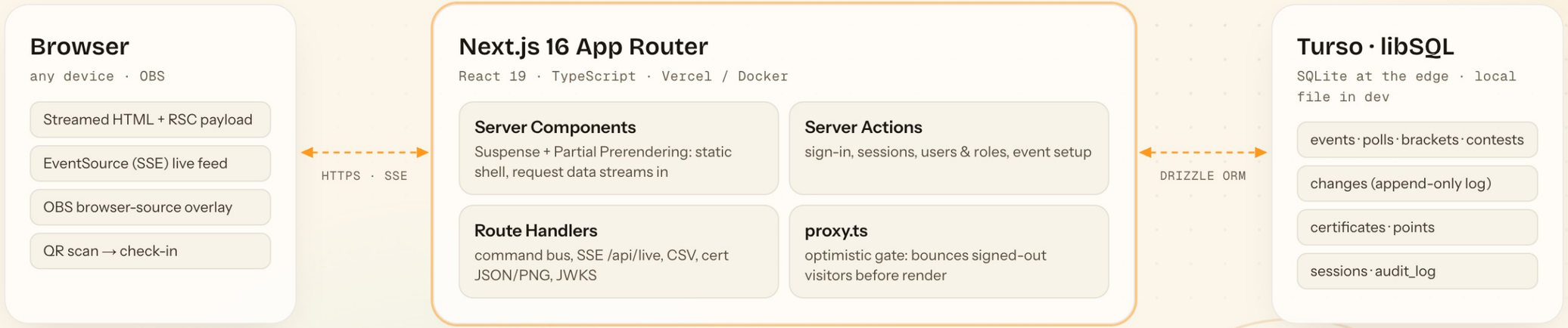
03

● RUBRIC 30% · TECHNICAL ARCHITECTURE

SYSTEM

A server-first Next.js 16 app on an edge database.

Pages ship a prerendered static shell; request-time data streams inside Suspense. Every write goes through one typed endpoint.



Cache Components / PPR · React 19 · TypeScript strict · Tailwind v4 · zod · Drizzle · jose · Vitest · GitHub Actions

04

RUBRIC 30% · TECHNICAL ARCHITECTURE

REALTIME

Realtime that survives serverless.

No websockets server, no lost messages: an append-only log in the database, tailed over SSE, resumable from the last event ID.

01
Typed command
POST /api/events/:id/commands — a discriminated union, one entry per feature

02
zod command bus
schema parse · house-scoped authz · DB-backed rate limit · audit

03
Append-only change log
INSERT INTO changes (seq) — redacted public view + full manager view

04
SSE stream
tails seq > cursor; resumes from Last-Event-ID, gap-free across instances

05
Pure reducer
useReducer folds each message into a snapshot — nothing to drift

src/lib/live.ts

```ts
// src/lib/live.ts — the log is the source of truth
export async function publish(
  channel: Channel | Channel[], type: string, payload: unknown,
) {
  const db = await getDb();
  const list = Array.isArray(channel) ? channel : [channel];
  const body = JSON.stringify(payload);
  await db.insert(schema.changes)
    .values(list.map((c) => ({ channel: c, type, payload: body })));
  for (const c of list) bus.emit("change", c);
}
```

- Correct across any number of instances — subscribers poll `seq > cursor`.
- In-process emitter wakes streams instantly on a single node.
- Public channels get redacted DTOs; managers get the full view.
- Streams close before platform timeouts; EventSource reconnects gap-free.

05

RUBRIC 30% · TECHNICAL ARCHITECTURE

SECURITY

Secure by construction, not by checklist.

scrypt passwords
Salted, timing-safe compare; throttled logins per account and network.

Revocable sessions
Server-side sessions; sign out other devices from the account page.

House-scoped RBAC
Organizers manage only events their house hosts; checked on every command.

CSP · HSTS
Strict Content-Security-Policy, HSTS preload, nosniff, frame-ancestors none.

Ed25519 certificates
Signed payloads + public JWKS; anyone can verify, revocation is checked live.

Audit log
Every sign-in, role change, moderation, score and wrap-up — CSV-safe export.

src/server/commands.ts

```ts
// src/server/commands.ts — one entry per feature
react: def({
  schema: z.object({ emoji: z.enum(REACTIONS) }),
  async run(ctx, { emoji }) {
    await limit(`react:${ctx.viewer.id}`, 8, 2, "Reacting");
    await publish(both(ctx.event.id), "reaction",
      { emoji, houseId: ctx.viewer.houseId });
  },
}),

export async function runCommand(
  db: DB, viewer: Viewer, eventId: string, raw: unknown,
) {
  …
  const manager = canManageEvent(viewer, event);
  if (cmd.manager && !manager) throw new HttpError(403, "Organizers only");
  const parsed = cmd.schema.safeParse(raw);
  if (!parsed.success)
    throw new HttpError(400, parsed.error.issues[0]?.message ?? "Invalid input");
  return cmd.run({ db, viewer, event }, parsed.data);
}
```

RUBRIC 30% · RELEVANCE & REAL-WORLD UTILITY

WHO IT'S FOR

Built around the three people in every house event.

Member

"I just watch a stream."

- Ask, upvote, vote and react live
- Predict bracket winners, back their house
- QR check-in counts toward points
- Certificate + portfolio, verifiable forever

Organizer

"Five tabs and a spreadsheet."

- One control room for the whole show
- Moderation queue, spotlight to stage
- Scores update bracket + overlay instantly
- Wrap-up issues every certificate in one click

Admin

"Who changed the points?"

- Users & roles with guardrails
- Audit log with filters + CSV
- Spotlight broadcast to every screen
- Revoke a certificate with a reason

● RUBRIC 30% · RELEVANCE & REAL-WORLD UTILITY

ADOPT IT

Any house can run it tomorrow — for free.

MODULAR STRUCTURE

```
src/
├ app/          routes · RSC pages · route handlers
├ server/       commands · queries · views · wrapup
├ lib/          auth · live · bracket · contest · certs
├ components/   live · admin · ui · site · three
├ db/           Drizzle schema · client · demo seed
└ proxy.ts      optimistic auth gate
drizzle/        SQL migrations
tests/          Vitest: engines + crypto
```

Pure domain engines (bracket, contest, achievements) have no I/O and are unit-tested.

Vercel + Turso

Both free tiers are enough for a house season.

One-command DB init

Migrations auto-apply on first boot; demo seed included.

Docker

One long-lived Node process — ideal for SSE.

.env.example

Every variable documented; secrets never committed.

MIT licensed

Fork it for any house, club or fest.

Demo mode

STAGECRAFT_DEMO=on seeds personas for judging; off for real use.

• RUBRIC 20% · UI/UX & PERFORMANCE

CRAFT

Fast, accessible, and a joy on a phone.

100	100	100
Accessibility	Best Practices	SEO

~110 fps scrolling

Parallax via CSS variables, no per-frame React renders.

Mobile-first

Mobile menu; event room reorders stream → interact.

WCAG contrast

Tokens tuned to ≥ 4.5:1 in both themes.

Keyboard + reduced motion

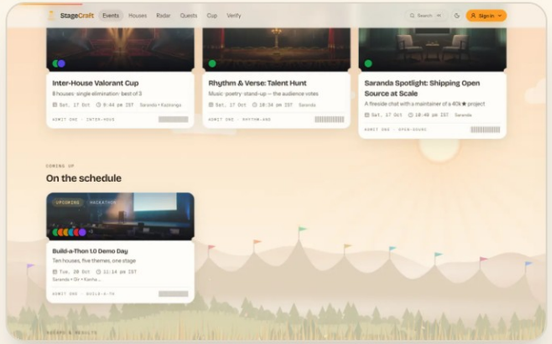
Skip link, visible focus, motion respects the OS.

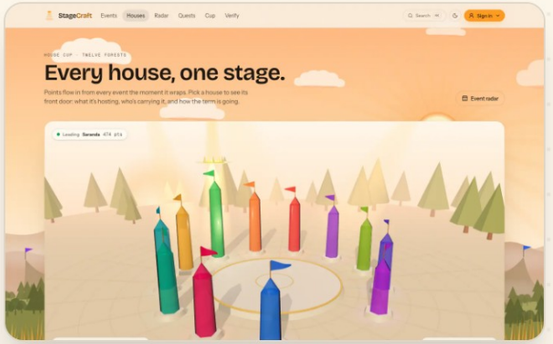
Daylight + Stage themes

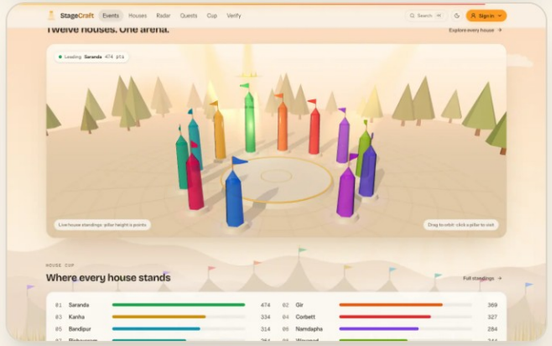
Paper by day; broadcast-black for the stage.

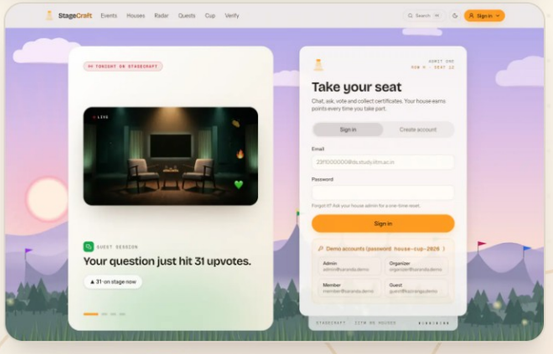
Living sky + 3D arena

Real IST sky; peaks are live house standings.





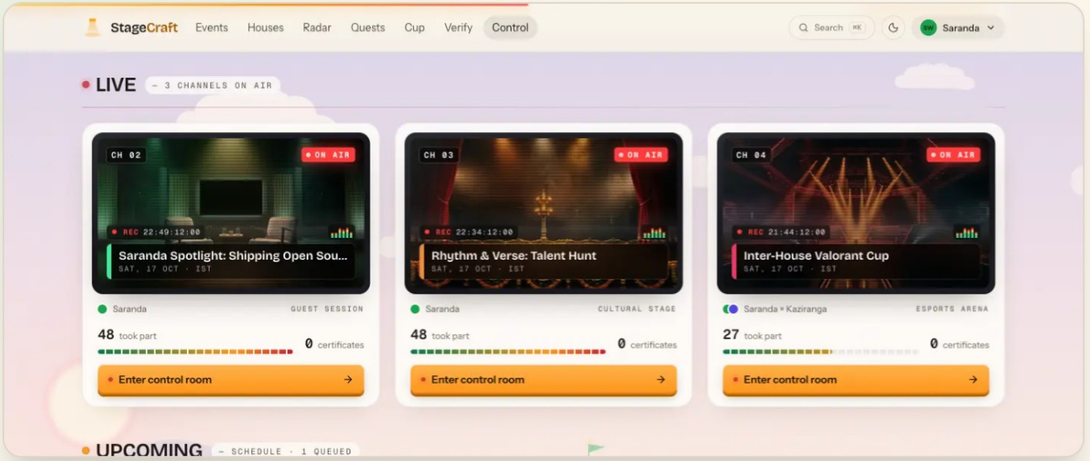




• RUBRIC 20% · ADMIN USABILITY & WORKFLOWS

RUN THE SHOW

Organizers drive. The app keeps score.



Control room
Segments, scenes, lower thirds, polls, scores.

Monitor wall
Stage, queue, chat and brackets at a glance.

Users & roles
Guardrails: no changing your own role or suspending yourself.

Audit log + CSV
Filter by action, actor, event, date.

Spotlight broadcast
Dims every screen, confetti in house colour.

Moderation
Approve, hide, pin, timed mute, slow mode.

WRAP-UP WORKFLOW

1

End the show
One toggle in the control room closes chat, votes and predictions.

2

Wrap up
Idempotent: podiums for every bracket and contest, speakers, organizers, engaged members.

3

Certificates
Ed25519-signed, on every profile, shareable PNG + public verify page.

4

House points
Awarded with dedupe keys; leaderboard and House Range update live.

PROOF

Shipped like production software.

36

Unit tests

Brackets, contests, achievements, canonical JSON, check-in, passwords

CI

Green on every push

Typecheck · lint · tests · build in GitHub Actions

0

Open high / medium issues

Independent security review — all high & medium findings fixed

0

Console errors

Across every page, both themes, desktop and mobile

- Lighthouse 100 a11y · best practices · SEO
- Strict CSP, no inline eval in production
- CSV export hardened against formula injection

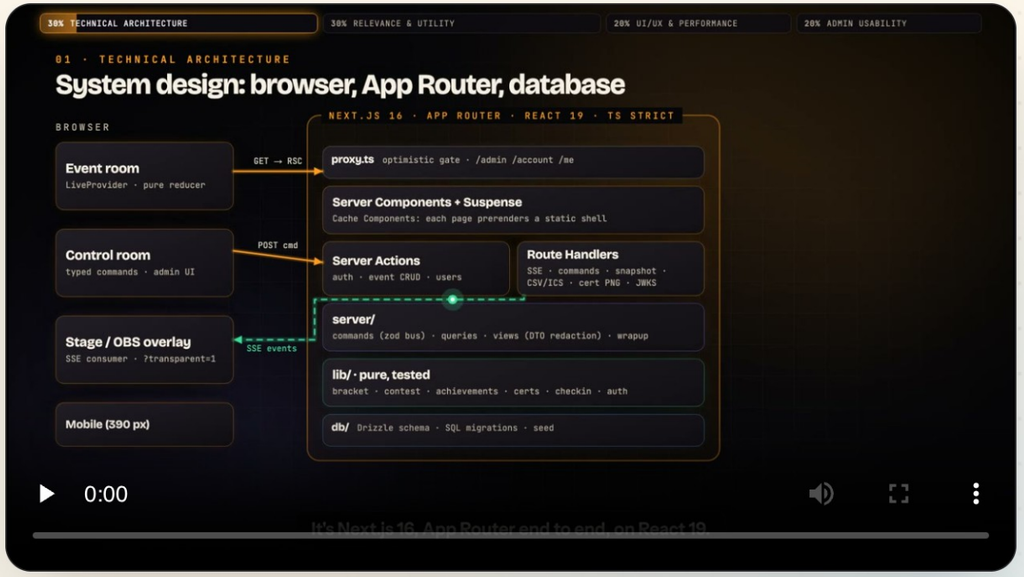
SEE IT RUN

Two short films.



Launch film

The product, from a member's seat to the wrap-up.



Technical pitch

Architecture, realtime pipeline and security, walked through.

STAGECRAFT · BUILD-A-THON 1.0

The audience is part of the show.

stagecraft-xi.vercel.app

github.com/lakshanmuruganandam/stagecraft

Password for every demo account: `house-cup-2026`

SCAN TO OPEN THE LIVE SITE